

BOARD OF INTERMEDIATE AND SECONDARY EDUCATION

Annual Examination 2026

Chemistry — SSC-I (9th Grade)

Syllabus: Chapter 10 — Acids, Bases and Salts

Total Marks: 65

Time Allowed: 2h 30m

Version: 1

General Instructions:

- Write your name, roll number and section clearly on the answer sheet.
- Section A is compulsory. Encircle the correct option on the question paper itself.
- In Section B, attempt **all 11** questions. For each question, attempt either the main question **OR** its alternative.
- In Section C, attempt **all 4** questions. For each question, attempt either the main question **OR** its alternative.
- Use black or blue ink only. Pencil is not allowed except for diagrams.

SECTION A — Multiple Choice Questions (12 × 1 = 12 Marks)

#	Question	A	B	C	D	Answer
1	According to the Arrhenius theory, an acid is a substance that ionizes in water to produce:	OH^- ions	H^+ ions	Na^+ ions	O^{2-} ions	ⒶⒷⒸⒹ
2	According to the Bronsted-Lowry theory, a base is a substance that:	donates a proton	donates OH^-	accepts a proton	produces H^+	ⒶⒷⒸⒹ
3	Which of the following cannot be classified as an Arrhenius acid?	HNO_3	H_2CO_3	CO_2	H_2SO_4	ⒶⒷⒸⒹ
4	In the reaction $\text{HCl} + \text{H}_2\text{O} \rightarrow \text{H}_3\text{O}^+ + \text{Cl}^-$, H_2O acts as a Bronsted-Lowry:	acid	base	neutral substance	salt	ⒶⒷⒸⒹ
5	An acid that ionizes completely in aqueous solution is called a:	weak acid	dilute acid	strong acid	concentrated acid	ⒶⒷⒸⒹ
6	Ethanoic acid (CH_3COOH) found in vinegar ionizes only about 5% in water. It is therefore classified as a:	strong acid	weak acid	strong base	weak base	ⒶⒷⒸⒹ
7	When zinc reacts with hydrochloric acid, the products are:	$\text{ZnO} + \text{H}_2$	$\text{ZnCl}_2 + \text{H}_2$	$\text{Zn}(\text{OH})_2 + \text{Cl}_2$	$\text{ZnCl}_2 + \text{O}_2$	ⒶⒷⒸⒹ
8	Milk of magnesia contains $\text{Mg}(\text{OH})_2$. It is used as an antacid to neutralize excess stomach acid (HCl). Which salt is formed in this reaction?	MgSO_4	MgCO_3	MgCl_2	MgO	ⒶⒷⒸⒹ
9	A base that is soluble in water is called an:	acid	salt	alkali	amphoteric substance	ⒶⒷⒸⒹ
10	Ammonia is a Bronsted-Lowry base because it:	contains OH^- group	ionizes to give OH^- ions	can accept a proton	can donate a proton	ⒶⒷⒸⒹ
11	Which of the following is an amphoteric oxide?	Na_2O	CaO	ZnO	MgO	ⒶⒷⒸⒹ
12	Acid rain is formed when acidic oxides such as SO_2 and NO_x dissolve in rainwater. Which acid is produced from dissolved SO_2 ?	HCl	HNO_3	$\text{H}_2\text{SO}_3/\text{H}_2\text{SO}_4$	H_3PO_4	ⒶⒷⒸⒹ

SECTION B — Short Questions (11 × 3 = 33 Marks)

Attempt all 11 questions. For each pair, attempt either the main question **OR** its alternative.

Part	Question	Marks	OR	OR Question	Marks
2(i)	State the Arrhenius definitions of an acid and a base. Write the ionization equations (with state symbols) for $\text{HCl}_{(\text{g})}$ and $\text{NaOH}_{(\text{s})}$ in water.	3	OR	State the Bronsted-Lowry definitions of an acid and a base. Identify the acid and base in: $\text{HNO}_3 + \text{H}_2\text{O} \rightarrow \text{H}_3\text{O}^+ + \text{NO}_3^-$.	3
2(ii)	Define a strong acid and a weak acid. Give two examples of each and explain why HCl is a strong acid while CH_3COOH is a weak acid.	3	OR	Define a strong base and a weak base. Give two examples of each. Write the ionization equation for NaOH in water using correct state symbols.	3
2(iii)	Write the ionic equation, with full charges and state symbols, for the neutralization of $\text{NaOH}_{(\text{aq})}$ by $\text{HCl}_{(\text{aq})}$. Identify the spectator ions and write the net ionic equation.	3	OR	Ammonium hydroxide reacts with nitric acid to form ammonium nitrate and water. Write the balanced chemical equation for this neutralization reaction.	3
2(iv)	List any three characteristic properties of acids, giving a chemical equation for each property.	3	OR	List any three characteristic properties of bases, giving a chemical equation for each property.	3
2(v)	Write balanced equations for the reaction of acids with metal carbonates. Give one example each for HCl reacting with Na_2CO_3 and H_2SO_4 reacting with CaCO_3 .	3	OR	Write balanced equations for the reaction of metals with acids. Give examples for (a) $\text{Zn} + \text{HCl}$ and (b) $\text{Mg} + \text{H}_2\text{SO}_4$. State the general pattern: $\text{Metal} + \text{Acid} \rightarrow ?$	3
2(vi)	What are alkalis? Give three examples of alkalis from Table 10.3 of your textbook. State two	3	OR	Describe how bases react with ammonium salts. Write the equations for (a) $\text{NaOH} + \text{NH}_4\text{Cl}$ and	3

	household products that contain alkalis.			(b) $\text{KOH} + \text{NH}_4\text{Cl}$.	
2(vii)	Define metallic oxides. Write equations showing how (a) sodium and (b) magnesium form metallic oxides. What happens when these oxides dissolve in water?	3	OR	What are amphoteric oxides and hydroxides? Name two examples of each from the textbook. Why are they called amphoteric?	3
2(viii)	In the reaction: $\text{H}_2\text{SO}_{4(l)} + \text{H}_2\text{O} \rightarrow 2\text{H}^+_{(aq)} + \text{SO}_4^{2-}_{(aq)}$, identify whether H_2SO_4 is a Bronsted acid or base, and whether it fully ionizes. Which category of acid does it belong to?	3	OR	In the reaction: $\text{H}_2\text{O} + \text{NH}_3 \rightleftharpoons \text{NH}_4^+ + \text{OH}^-$, identify the Bronsted-Lowry acid and base. Explain why water is described as amphoteric.	3
2(ix)	State five general properties of acids from Section 10.3 of your textbook (taste, litmus, conductivity, reaction with metals, reaction with bases).	3	OR	State five general properties of bases from Section 10.4 of your textbook (taste, litmus, touch, neutralization, reaction with ammonium salts).	3
2(x)	Define acid rain. State two main acidic gases responsible for acid rain. Describe two damaging effects of acid rain on structures and materials.	3	OR	Explain why normal rain is slightly acidic. How does acid rain differ from normal rain? State one biological and one structural effect of acid rain.	3
2(xi)	Compare HCl , HNO_3 , and H_2SO_4 under the headings: formula, common use (one use each), and whether each is a strong or weak acid.	3	OR	Compare NaOH , KOH , and Ca(OH)_2 under the headings: formula, common use (one use each), and whether each is a strong or weak base.	3

SECTION C — Long Questions (4 × 5 = 20 Marks) — Attempt all 4

Part	Question	Marks	OR	OR Question	Marks
3(i)	(a) Compare the Arrhenius and Bronsted-Lowry theories of acids and bases. State one limitation of the Arrhenius theory that the Bronsted-Lowry theory overcomes. [3] (b) Classify each of the following as a Bronsted-Lowry acid or base. Justify each answer: (i) HCl in $\text{HCl} + \text{H}_2\text{O} \rightarrow \text{H}_3\text{O}^+ + \text{Cl}^-$ (ii) NH_3 in $\text{H}_2\text{O} + \text{NH}_3 \rightleftharpoons \text{NH}_4^+ + \text{OH}^-$. [2]	5	OR	(a) What is meant by saying water is amphoteric? Write two equations from the textbook showing water acting once as a Bronsted acid and once as a Bronsted base. [3] (b) Identify the Bronsted acids and bases in each of the following reactions from Exercise 7 of the textbook: (i) $\text{CH}_3\text{COOH}_{(aq)} + \text{H}_2\text{O} \rightleftharpoons \text{CH}_3\text{COO}^-_{(aq)} + \text{H}_3\text{O}^+_{(aq)}$ (ii) $\text{HCO}_3^-_{(g)} + \text{H}_2\text{O}_{(l)} \rightleftharpoons \text{CO}_3^{2-}_{(aq)} + \text{H}_3\text{O}^+_{(aq)}$. [2]	5
3(ii)	(a) Describe the characteristic properties of acids in terms of their reactions with: (i) metals (ii) metal carbonates (iii) bases (neutralization). Write a balanced chemical equation for each. [3] (b) Write the full ionic equation with charges and state symbols for $\text{HCl}_{(aq)} + \text{NaOH}_{(aq)} \rightarrow \text{NaCl}_{(aq)} + \text{H}_2\text{O}_{(l)}$. Identify the spectator ions and state the net ionic equation. Explain why all strong acid-strong base neutralizations share the same net ionic equation. [2]	5	OR	(a) Describe the characteristic properties of bases in terms of their reactions with: (i) acids (neutralization) (ii) ammonium salts. Write a balanced equation for each. [3] (b) Write equations for the reactions of: (i) $\text{H}_2\text{SO}_4 + \text{Mg(OH)}_2$ and (ii) $\text{H}_2\text{SO}_4 + \text{Na}_2\text{CO}_3$. For reaction (ii), identify all three products and explain why bubbling is observed. [2]	5
3(iii)	(a) Define oxides and hydroxides of metals. Explain the difference between a basic oxide and an amphoteric oxide with two examples of each. [3] (b) Write equations showing: (i) how Na_2O reacts with water to form NaOH and (ii) how CaO reacts with water to form Ca(OH)_2 . State whether the products are strong or weak bases and give one use of each. [2]	5	OR	(a) Explain, using equations, why metallic oxides are basic in nature. Show what happens when the oxide ion (O^{2-}) reacts with water ($\text{O}^{2-} + \text{H-OH} \rightarrow 2\text{OH}^-$). [3] (b) Define amphoteric substances. Water is amphoteric. Using the two reactions from Section 10.1.2, show water acting as both acid and base. State one other amphoteric oxide and one amphoteric hydroxide from the textbook. [2]	5
3(iv)	(a) Define acid rain. List the main acidic gases produced by human activity that cause acid rain and explain how they form acidic solutions in rainwater. [3] (b) Describe any four damaging effects of acid rain. Your answer should include effects on: (i) metal structures (ii) stone buildings/statues containing CaCO_3 (iii) biological systems. Write a chemical equation to illustrate one of these effects. [2]	5	OR	(a) Explain the properties of acids that make acid rain corrosive. Use Table 10.1 from the textbook as a starting point. Why do acids react with metals and metal carbonates found in buildings? [3] (b) Normal rainwater has a pH below 7 even without industrial pollution. Explain why this is so. Contrast the pH of normal rain with acid rain. Give one example of the long-term environmental consequence of acid rain on (i) water bodies and (ii) soil. [2]	5

Invigilator's Signature: _____

Candidate's Signature: _____

Chapters Covered: Chapter 10 — Acids, Bases and Salts (pp. 141-149)

— End of Question Paper —